

Validation of the Prognostic Usefulness of the Gene Expression Profiling Test in Patients with Uveal Melanoma

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Purpose: To validate the prognostic usefulness of gene expression profile (GEP) testing in patients with uveal melanoma. To determine whether combining tumor size with the GEP classification provides additional prognostic value.

Design: Retrospective analysis.

Participants: Patients with a diagnosis of choroidal melanoma examined at Yale New Haven Hospital; University of California, San Diego; and Memorial Sloan Kettering Cancer Center.

Methods: Patients' demographic and clinical data and tumor characteristics were collected. Univariate and multivariate Cox hazard regression analysis were used to assess the association between tumor characteristics and GEP classification with metastasis as an outcome.

Main Outcome Measures: Metastasis-free survival (MFS).

Results: Of the 337 individuals included in the study, 87 demonstrated metastases. The mean follow-up time was 37.2 (standard deviation [SD], 40.2) months for patients with metastases and 55.0 (SD, 49.3) months for those without metastases. Tumors of larger thickness and GEP class 2 (vs. class 1) were associated significantly with increased risk of metastasis. Tumor thickness showed better prognostic usefulness than GEP classification (Wald statistic, 40.7 and 24.2, respectively). Class 2 tumors with a thickness of 7.0 mm or more were associated with increased risk of metastasis than tumors with a thickness of < 7.0 mm (hazard ratio [HR], 3.23; 95% confidence interval [CI], 1.61–6.51), whereas class 1 tumors with a thickness of 9.0 mm or more were associated with increased risk of metastasis than tumors with a thickness of < 9.0 mm (HR, 2.07; 95% CI, 0.86–4.99). No difference in MFS was found between patients with class 1A tumors compared with those with class 1B tumors ($P = 0.8$). Patients with class 2 tumors showed an observed 5-year MFS of 47.5% (95% CI, 36.0%–62.8%).

Conclusions: Tumor size was the most significant predictor of metastasis and provided additional prognostic value independent of GEP classification. In addition, rates of metastasis for class 2 tumors were lower than estimates reported by Castle Bioscience, and no difference in rates of metastasis were found between class 1A and 1B tumors. This indicates that tumor size should be accounted for when relying on GEP for prognostication and that patients with GEP class 1A or 1B tumors may benefit from the same metastatic surveillance protocols.

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Uveal melanoma (UM) is the most common primary intraocular cancer in adults. It arises from melanocytes within the uveal tract (choroid, ciliary body, and iris).^{1–3} Approximately one third of patients with UM will demonstrate metastatic disease within 5 years, which usually is fatal within 1 year of clinical onset.^{2,4} Uveal melanomas that tend to metastasize often have cells of epithelioid morphologic features with enlarged nuclei, and they frequently harbor mutations in the *BAP1* (BRCA1-associated protein-1) gene or exhibit genomic copy loss of chromosome 3.^{1,5,6}

Additionally, highly metastatic tumors can be distinguished from their indolent counterparts by their gene expression profile (GEP).^{1,7,8} In clinical practice, biopsy samples are obtained commonly at the time of treatment and are analyzed to assign a prognostic class to the tumor with the goal of providing an accurate estimate of the metastatic risk. Two main molecular prognostic tests are commercially available and used commonly. The first is a DNA-based test (offered by Impact Genetics) that analyzes the tumor's chromosomal number alterations,